Build a complete, production-ready, free all-in-one QR Code Generator SaaS website named QRNova.

GOAL: Launch with ZERO infrastructure cost. Must be deployable on Cloudflare Pages free tier. Use 100% client-side QR generation (use qrcode.js or qr-code-styling library). NO database, NO paid backend for basic generation.

1. CORE FEATURES:

Build 21+ QR tools in one page: URL, Text, WiFi, Email, Phone, SMS, WhatsApp, vCard/Contact, Google Maps/Location, Event, Social Media, Instagram, Facebook, YouTube, LinkedIn, X/Twitter, App Download, PDF, Menu, Business, Bitcoin.

Flow: Choose QR Type -> Enter Info -> Customize -> Live Preview -> Download. No signup required for basic use.

2. CUSTOMIZATION:

Foreground color, background color, gradient, dot style, eye/corner style, size, margin, error correction (L,M,Q,H), logo upload with size control, transparent background. Show warning if contrast is low: "This design may be difficult to scan..."

3. DOWNLOAD:

Buttons: Download PNG, Download JPG, Download SVG. High-res 1024px+ support. Must work client-side.

4. DESIGN - VERY IMPORTANT:

Premium modern AI SaaS look. NOT old QR site. Colors: White, off-white, #f8f9ff, soft blue, lavender, light purple. Use rounded-2xl cards, soft shadows, glassmorphism, clean typography (Inter/Sora). Minimal, fast, futuristic. Mobile-first.

Desktop Layout: Left = Controls, Right = Sticky Live QR Preview + Download.

Mobile: Vertical stack.

5. HOMEPAGE SEO:

Hero Badge: FREE QR CODE GENERATOR

H1: Free QR Code Generator – Create QR Codes Instantly

Subtext: Create free QR codes for websites, WiFi, WhatsApp, social media, contact details, menus and more. Customize and download instantly. No signup required.

CTA: Create QR Code + Explore QR Tools. Show working generator on homepage itself.

6. SEO PAGES - Create separate pages:

/ , /url-qr-code-generator, /wifi-qr-code-generator, /whatsapp-qr-code-generator, /vcard-qr-code-generator, /email-qr-code-generator, /google-maps-qr-code-generator, /instagram-qr-code-generator, /menu-qr-code-generator, /business-qr-code-generator etc.

Each page: Unique Title, Meta Description, H1, Intro, Working Generator, How-to steps, Use-cases, FAQ with FAQPage schema, Breadcrumbs, Canonical, OG tags.

7. TECHNICAL SEO:

Semantic HTML, one H1 per page, fast Core Web Vitals, generate sitemap.xml and robots.txt, implement schemas: WebSite, SoftwareApplication, BreadcrumbList, FAQPage. No fake reviews.

8. CLOUDFLARE PAGES READY:

Framework: React + Vite. Build command: npm run build. Output dir: dist. No server dependencies. Static only. So it works on yourbrand.pages.dev without modification.

9. ADSTERRA READY:

Create reusable component <AdSlot id="..."/>. Place ad slots: Below generator, Between content sections, Sidebar on desktop, Inside blog. NEVER place ads over generator, preview, or download buttons. Keep UX clean. Do not hardcode ads, just placeholder divs.

10. OTHER PAGES:

Blog (12+ articles like What is QR Code, How to create WiFi QR etc), Privacy Policy, Terms, Cookie Policy, Contact, About, Footer with internal links.

11. FINAL CHECKLIST:

All 21 QR types working, customization working, logo upload working, PNG/JPG/SVG download working, mobile responsive, no signup wall, fast, no console errors, sitemap.xml + robots.txt exists, deployable to Cloudflare Pages.

Build as REAL FUNCTIONAL PRODUCT, not just desi